

Lab 5, Act 1: Answer Key

Finding the Peak

NRES 746, Instructor Answer Key (not for student distribution)

Each item repeats the handout's question, then gives the answer, a short rationale, and mistakes to watch for. Questions with one correct answer are answered directly. Open-ended items (1c, the “does it make sense” half of 2c, and 3a) get discussion points instead. All numbers are computed in R when this file is knitted.

Data: detections $k = (3, 2, 7)$ out of $N = 10$ visits at each of three wetlands.

Part 1: The likelihood

1a.

Write the likelihood of all three wetlands' data, $\mathcal{L}_3(p)$, using product notation.

Answer.

$$\mathcal{L}_3(p) = \prod_{i=1}^3 \binom{10}{k_i} p^{k_i} (1-p)^{10-k_i} = \binom{10}{3} \binom{10}{2} \binom{10}{7} p^{12} (1-p)^{18}$$

Why. Each wetland's count is binomial with $N = 10$. Independence means the joint probability is the product. The exponents add: $3 + 2 + 7 = 12$ detections and $7 + 8 + 3 = 18$ misses.

Watch for.

- Treating each wetland as a single Bernoulli trial (0/1) instead of a binomial count out of 10.
- Adding the three probabilities instead of multiplying.
- Writing the product with no subscript i , so all three terms look identical.

1b.

Take the log and simplify $\ell_3(p) = \log \mathcal{L}_3(p)$ into a sum, as far as you can.

Answer.

$$\ell_3(p) = \sum_{i=1}^3 \left[\log \binom{10}{k_i} + k_i \log p + (10 - k_i) \log(1 - p) \right] = C + 12 \log p + 18 \log(1 - p)$$

where $C = \log \binom{10}{3} + \log \binom{10}{2} + \log \binom{10}{7} = \log(120 \cdot 45 \cdot 120) \approx 13.38$.

Why. $\log(uv) = \log u + \log v$ turns the product into a sum, and $\log(u^c) = c \log u$ brings the exponents down.

Watch for.

- Writing $\log(1 - p)$ as $\log 1 - \log p$.
- Losing the $(10 - k_i)$ exponent, e.g. writing $\log(1 - p)$ with no coefficient.

1c.

Your answer to 1b includes binomial coefficients such as $\binom{10}{3}$. If you left them out, would that change which value of p maximizes $\ell_3(p)$? Why or why not?

Answer. No.

Why. The binomial coefficients don't depend on p . In the log scale they add a constant, which shifts the whole curve up or down without moving its peak. (On the likelihood scale they multiply the curve by a constant, which also leaves the peak in place.) Check:

```
optimize(ll_3, c(0, 1), maximum = TRUE)$maximum
```

```
## [1] 0.4000015
```

```
optimize(ll_3_nchoose, c(0, 1), maximum = TRUE)$maximum
```

```
## [1] 0.4000015
```

Watch for. “Yes, because the likelihood value changes.” The height changes, the location of the peak does not.

Part 2: Find the peak

2a.

Find the derivative with respect to p of either $\mathcal{L}_3(p)$ or $\ell_3(p)$, your choice (both peak at the same p). In general, the log-likelihood is easier to differentiate.

Answer (log version).

$$\frac{d\ell_3}{dp} = \frac{12}{p} - \frac{18}{1-p}$$

Answer (raw version). With $K = \binom{10}{3}\binom{10}{2}\binom{10}{7}$,

$$\frac{d\mathcal{L}_3}{dp} = K [12p^{11}(1-p)^{18} - 18p^{12}(1-p)^{17}] = K p^{11}(1-p)^{17} [12(1-p) - 18p]$$

Why. Log version: $\frac{d}{dp} \log p = \frac{1}{p}$, and for $\log(1-p)$ the chain rule gives $\frac{1}{1-p} \cdot (-1)$. The constant drops out. Raw version: product rule on $p^{12}(1-p)^{18}$, with the chain rule on $(1-p)^{18}$, then factor out $p^{11}(1-p)^{17}$. Both change sign at the same p (positive below 0.4, zero at 0.4, negative above):

```
K <- prod(choose(N, k))
d_raw <- function(p) K * (12 * p^11 * (1 - p)^18 - 18 * p^12 * (1 - p)^17)
d_log <- function(p) 12 / p - 18 / (1 - p)
p_check <- c(0.3, 0.4, 0.5)
rbind(raw = signif(d_raw(p_check), 3), log = round(d_log(p_check), 3))
```

```
##           [,1]      [,2]      [,3]
## raw  0.00801 8.58e-18 -0.00724
## log 14.28600 0.00e+00 -12.00000
```

Watch for.

- **The missing minus sign** from the chain rule on $\log(1-p)$ or $(1-p)^{18}$. This is the most common error in either version, and it leads to an equation with no solution between 0 and 1.
- Raw version: not factoring before solving, which leaves a high-degree polynomial that looks hopeless.
- Differentiating $\binom{10}{k_i}$ as if it depended on p .

2b.

Set your derivative equal to zero and solve for $\hat{p}$, the maximum likelihood estimate. Try to do this on your own first, but you can get help by looking at the Bolker book (beginning of Ch. 6).

Answer. $\hat{p} = 0.4$.

Why.

$$\frac{12}{p} = \frac{18}{1-p} \Rightarrow 12(1-p) = 18p \Rightarrow 12 = 30p \Rightarrow \hat{p} = \frac{12}{30} = 0.4$$

The raw version reaches the same bracket, $12(1-p) - 18p = 0$. The second derivative of ℓ_3 , $-12/p^2 - 18/(1-p)^2$, is negative for every p , so this is a maximum (students weren't asked to check).

Watch for.

- Raw version: also reporting $p = 0$ and $p = 1$, where $p^{11}(1-p)^{17} = 0$. There the likelihood is 0, the minimum, not the peak.
- Cross-multiplying incorrectly (e.g. $12p = 18(1-p)$, giving 0.6).
- Averaging the three wetlands' proportions separately. That also gives 0.4 here only because all N_i are equal (see 2c).

2c.

Now suppose you surveyed n wetlands, each visited N times, with k_i detections at wetland i . Write a general formula for $\hat{p}$. Does it make sense?

Answer.

$$\hat{p} = \frac{\sum_{i=1}^n k_i}{nN}$$

Total detections divided by total visits.

Why (discussion). Repeat 2a-2b with $\sum k_i$ in place of 12 and $\sum(N - k_i) = nN - \sum k_i$ in place of 18. It makes sense: the best estimate of a per-visit detection rate is the observed fraction of visits with a detection. Students may notice this is what they would have guessed without any calculus, which is a good sign rather than a letdown. The likelihood machinery earns its keep when there is no intuitive answer (Act 2).

Watch for. Writing $\frac{1}{n} \sum \frac{k_i}{N}$ and calling it different. With equal N it is the same formula. (It would differ if wetlands had different numbers of visits, where the MLE is $\sum k_i / \sum N_i$.)

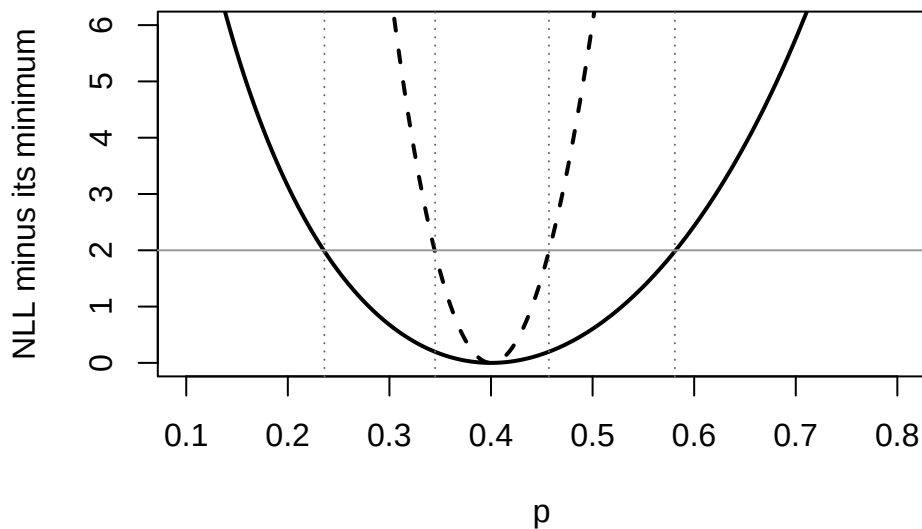
Part 3: How sure are we?

3a.

A colleague ran the same survey at 30 spring complexes where the species is known to be present (10 visits each) and heard the frog on 120 of the 300 visits. [Figure of both NLL curves, each shifted so its minimum is zero.] Do the two surveys give the same $\hat{p}$? Which survey pins p down more tightly, and what feature of the curve shows it?

Answer. Yes, both give $\hat{p} = 0.4$ ($120/300 = 12/30$). The 30-wetland survey pins p down more tightly: its curve is much narrower and steeper around the minimum.

Why (discussion). A steep curve means the NLL rises quickly as p moves away from 0.4, so values far from 0.4 are much less supported by the data. A shallow curve means a wide range of p values fit nearly as well. For reference, the range of p within 2 NLL units of the minimum is about 0.236 to 0.581 for 3 wetlands and 0.345 to 0.457 for 30 wetlands. (Don't give students this "within 2 units" rule yet. It comes later in the semester.)



Watch for.

- Saying the 30-wetland survey gives a “better” or “higher” $\hat{p}$. The estimate is identical, only the precision differs.
- Reading the height of the curves as meaningful. Both were shifted to a minimum of zero, and only the shape matters here.

Group discussion and revision

No fixed answer. The derivative in 2a (the sign on the $\log(1 - p)$ term) and the difference between “same estimate” and “same certainty” in 3a are the most likely sources of disagreement. Muddiest points worth noting for the next lecture: likelihood vs. probability, and whether a narrow curve means the estimate is “right.”